Trial SET #7 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only **60** minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

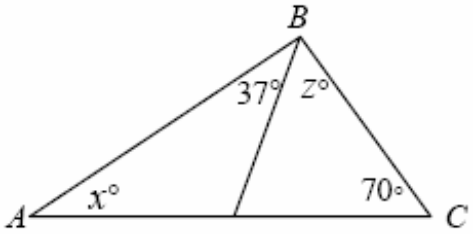
1. Chandler is building a fence in the following method: He grounds 25 poles, each 16 cm thick, in 1.5 meters spaces from each other. He then connects the poles with a barbed wire. What is the total length of the fence?

A. 40.5; B. 37.5; C. 38.5; D. 41.5; E. 40;

2. The age of the Earth is approximately 1.3×10^{17} seconds, and one year is approximately 3.2×10^{7} seconds. Which of the following is closest to the age of the Earth in years?
A. 2.5×10^{9}
B. 4.1×10^{9}
C. 1.9×10^{10}
D. 2.5×10^{11}
E. 4.1×10^{11}

3. When an integer Q is divided by 6, the remainder is 4. Which of the following is not a multiple of 6?

A. Q + 2; B. Q - 4 ; C. 2Q + 6; D. 3Q; E. 4Q + 2;



4. In ΔABC above, what is x in terms of z ?

(A) $z + 73$ (B) $z - 73$ (C) $70 - z$ (D) $z - 70$ (E) $73 - z$

5. A bag contains 25 slips of paper, on each of which a different integer from 1 to 25 is written. Blindfolded, Scott draws one of the slips of paper. He wins if the number on the slip he draws is a multiple of 3 or 5. What is the probability that Scott wins?

A. 1/25; B. 8/25; C. 11/25; D. 12/25; E. 13/25;

6. $\left(3^9-3^8\right)\left(3^7-3^6\right)=$

A. 3^4 B. 3^{14} C. 6^{14} D. 2×3^{14} E. 4×3^{14}

7. $\left(\sqrt[4]{\frac{ab}{\sqrt{ab}}}\right)^2=?$

A. ab; B. $(ab)^{-\frac{1}{2}}$; C. $(ab)^{-\frac{3}{4}}$; D. $(ab)^{\frac{3}{4}}$; E. None of these;

8. The three-digit integer *ABC* is formed from the digits *A,B,* and *C*, where *A=ab, B=bc,* and *C = ac*. If *a* is an element of [1, 3], *b* is an element of [2, 3], and *c* is an element of [1, 2], then which of the following numbers CANNOT be *ABC*?

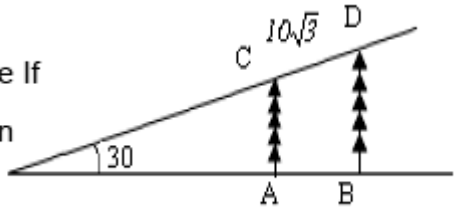
A. 221; B. 222; C. 214; D. 322; E. 646;

Model	Number Sold (1000's)
A	⊕⊕⊕⊕⊕⊕⊕⊕⊕
B	⊕⊕⊕⊕⊕⊕⊕⊕⊕⊕

9. If the selling price of model *B* is 60% more than the selling price of model *A*, what percent of the total sales do the sales of model *A* represent?

A. 25%; B. 36%; C. 40%; D. 50%; E. 60%;

10. In the given figure If $|CD| = 10\sqrt{3}$ cm then find the distance between tree A and tree B.

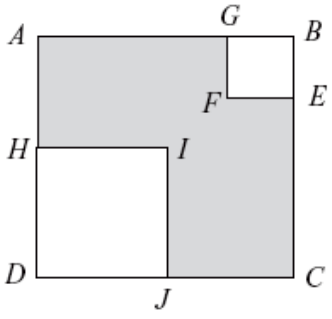


A) $10\sqrt{3}$ B) $15\sqrt{3}$ C) 15 D) 30 E) 45

11. Let *P* and *Q* be points 2 inches apart, and let *A* be the area, in square inches, of a circle that passes through *P* and *Q*. Which of the following is the set of all possible values of *A*?

A. $0 < A$; B. $0 < A \leq \pi$; C. $A = \pi$; D. $A > \pi$; E. $A \geq \pi$;

12. In the figure above, *ABCD*, *BEFG*, and *DHIJ* are squares with *AB* = 2(*DH*) and *DH* = 2(*BE*). If a point is chosen at random inside square *ABCD*, what is the probability it will be in the shaded region?



A. 1/3; B. 1/2; C. 5/8; D. 11/16; E. 3/4;

13. $1/2^{12}+2/2^{13}+3/2^{14}+4/2^{15}=$

A. $(1/2)^{10}$; B. $(1/2)^{12}$; C. $15/2^{15}$; D. $2/2^{10}$; E. $13/2^{14}$;

14. Of the n students in a certain class, x took a course in biology and y took a course in history, and z took neither of the courses. If there were students who took both two courses, in terms of n, x, y, and z, what fraction of the students took both two courses?

(A) $1-\dfrac{x+y+z}{n}$

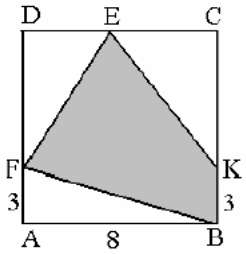
(B) $\dfrac{x+y+z}{n}-1$

(C) $\dfrac{x+y-z-n}{n}$

(D) $\dfrac{n+x+y-z}{n}$

(E) $\dfrac{n-x-y+z}{n}$

15. ABCD is a square, $|FA|=|KB|=3$ cm and $|AB|=8$ cm then find $A(EFBK)$.



A) 24 B) 28 C) 32 D) 36 E) 42

16. Get a number between $3-2\sqrt{2}$ and $3+2\sqrt{2}$ so that these three numbers form a geometric sequence.

A. 1 B. 2 C. 3 D. $\sqrt{2}$ E. $\sqrt{2}+1$

17. Find the smallest acute angle x, that satisfies the equality $\sin(5x) = \cos 35^0$.

A. 7; B. 11; C. 18; D. 24; E. 35;

18. If $a = \frac{1}{3}, b = \frac{1}{10}, c = \frac{2}{3}$, then $\sqrt[4]{\frac{16a^8b^4}{0{,}0001.c^{12}}}$ =?

A. 4/3; B. 1/2; C. 3/8; D. 3/4; E. 2/3;

19. If $a = \log_2 7$, $b = \log_5 24$ and $c = \log_3 50$ are given. Which of the following is true?

A) a < b < c B) c < a < b C) b < a < c
D) b < c < a E) a < c < b

AMOUNT OF BACTERIA PRESENT

Time	Amount
1:00 P.M.	10.0 grams
4:00 P.M.	x grams
7:00 P.M.	14.4 grams

20. Data for a certain biology experiment are given in the table above. If the amount of bacteria present increased by the same fraction during each of the two 3-hour periods shown, how many grams of bacteria were present at 4:00 P.M.?

A. 12.0
B. 12.1
C. 12.2
D. 12.3
E. 12.4

21. Find the derivative of the function $f(x)=(x^2+1)/(x+1)$ at x = 1.

A. 1; B. 0; C. 0,5; D. –1; E. –0,5;

22. $\int_0^1 (e^x - x)dx = ?$

A. e^2; B. $\frac{e-2}{3}$; C. $\frac{e+2}{3}$; D. $\frac{e-3}{4}$; E. $\frac{2e-3}{2}$;

23. Sandy has a collection of 10 different books that includes 7 same size small and 3 same size large books. The size of 1 large book is equal to the size of 3 small books. In how many ways can she select books to her trip if she has space available for 1 large and 4 small books?
(A) 35
(B) 56
(C) 126
(D) 127
(E) 252

24. $\frac{\sin^2 x}{\cos x - \sec x} + \frac{\cos^2 x}{\sin x - \csc x} = ?$

A) –sinx – cosx B) sinx + cosx C) $1 - 3\sin^2 x$
D) $2\cos^2 x + 3$ E)1

25. Which one is not equal to $\cos\left(\frac{\pi}{2}+x\right)$.

a) $\sin(-x)$ b) $\cos(\frac{3\pi}{2} - x)$ c) $\sin(\pi+x)$ d) $\sin(x - 2\pi)$ e) $-\cos(x - \frac{\pi}{2})$

26. How many zeroes does 50! have as last digits when written in decimal form?

A. 12; B. 14; C. 16; D. 22; E. the last digit is not zero;

27. For any integer k greater than 1, the symbol k^* denotes the product of all the fractions of the form $1/t$, where t is an integer between 1 and k, inclusive. What is the value of $5^*/4^*$?
A. 5
B. 5/4
C. 4/5
D. 1/4
E. 1/5

28. If $x-2 \le z \le x+5$, where does the value of x lie in terms of z ?
A. $z-5 \le x \le z$
B. $z-5 \le x \le z+2$
C. $z-2 \le x \le z+5$
D. $z-2 \le x \le z$
E. Other

29. What are the sings of $\sin(181^0)$, $\cos(-91^0)$, $\tan(-275^0)$ and $\cot(350^0)$, respectively?

A. -,+,+,- B. -,-,+,- C. -,+,-,- D. -,-,-,- E. -,+,-,+

30. A thin piece of wire 40 meters long is cut into two pieces. One piece is used to form a circle with radius r, and the other is used to form a square. No wire is left over. Which of the following represents the total area, in square meters, of the circular and the square regions in terms of r?

A. πr^2 B. $\pi r^2 + 10$ C. $\pi r^2 + \frac{1}{4}\ \pi^2 r^2$
D. $\pi r^2 + (40-2\pi r)^2$ E. $\pi r^2 + (10-\frac{1}{2}\ \pi r)^2$